

Day 05 - Exploring Linux Fundamentals in Depth

Objective

To deepen my understanding of Linux by exploring its architecture, file system, package management, remote administration concepts, and process management. These concepts are essential for system administration, Cloud Computing, and DevOps.

Desktop Environment in Linux

A **Desktop Environment (DE)** provides the graphical user interface (GUI) that allows users to interact with the Linux operating system easily.

Popular Desktop Environments

- GNOME
- KDE Plasma
- XFCE
- Cinnamon
- MATE

Components of a Desktop Environment

- Desktop
- Window Manager
- File Manager
- Application Launcher
- Taskbar
- Settings Manager

Although Linux servers often run **without a GUI**, desktop environments are useful for learning and personal systems.

Software Repositories & Package Management

Linux software is installed using **package managers**, which download applications from trusted **repositories**.

What is a Repository?

A repository is an online collection of software packages maintained by Linux distributions.

Common Package Managers

Distribution	Package Manager
Ubuntu / Debian	APT
CentOS / RHEL	YUM / DNF
Arch Linux	Pacman

Common Commands

Update package list:

```
sudo apt update
```

Upgrade installed packages:

```
sudo apt upgrade
```

Install a package:

```
sudo apt install package-name
```

Remove a package:

```
sudo apt remove package-name
```

Remote Access & Server Management

Cloud servers are usually managed remotely instead of physically accessing them.

Common Remote Access Tools

- SSH (Secure Shell)
- PuTTY
- Remote Desktop (for GUI systems)
- WinSCP (File Transfer)

Benefits

- Secure administration
- Remote troubleshooting
- File transfer
- Server monitoring
- Automation

SSH is the most commonly used tool by Linux administrators and DevOps engineers.

Linux System Architecture

Linux follows a layered architecture.

```
+-----+
| User Applications |
+-----+
| Shell             |
+-----+
```

```
| System Utilities |  
+-----+  
| Linux Kernel   |  
+-----+  
| Hardware       |  
+-----+
```

Hardware

Physical components such as CPU, RAM, storage devices, and network adapters.

Kernel

The core of Linux.

Responsibilities:

- Memory Management
- Process Management
- Device Management
- File System Management
- Security

Shell

The command interpreter that allows users to interact with the operating system.

Examples:

- Bash
- Zsh
- Fish

User Applications

Programs such as:

- Firefox
- VS Code
- Docker

- Git

Hardware Information in Linux

Linux provides several commands to view hardware details.

Examples:

Display CPU information

```
lscpu
```

Display memory information

```
free -h
```

Display storage devices

```
lsblk
```

Display kernel information

```
uname -a
```

Linux File System Structure

Linux follows a hierarchical file system starting from the **root directory (/)**.

Important directories:

Directory	Purpose
/	Root directory
/home	User files
/bin	Essential commands
/etc	Configuration files

Directory	Purpose
/var	Logs and variable data
/usr	Installed applications
/tmp	Temporary files
/dev	Device files

Linux vs Unix

Linux	Unix
Open Source	Mostly Proprietary
Free	Commercial Licenses
Community Developed	Vendor Developed
Runs on many devices	Primarily Enterprise Systems

Similarities

- Multi-user
- Multitasking
- Command-line interface
- Similar file system structure
- Security features

Process States in Linux

Every running program is called a **process**.

A process can exist in different states.

Running (R)

Currently executing on the CPU.

Sleeping (S)

Waiting for an event or resource.

Waiting (D)

Waiting for input/output operations.

Stopped (T)

Execution has been paused.

Zombie (Z)

The process has completed execution but still has an entry in the process table until its parent process collects its exit status.

Hands-On Practice

- Explored Linux directory structure
 - Viewed hardware information
 - Learned package management concepts
 - Studied Linux architecture
 - Understood process lifecycle
 - Explored remote server management concepts
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Key Learnings

- Linux architecture consists of Hardware, Kernel, Shell, System Utilities, and User Applications.
- Linux repositories provide secure software installation.
- Most cloud servers are managed remotely using SSH.
- Linux uses a hierarchical file system starting from the root directory.
- Linux and Unix share many concepts but differ in licensing and development.

- Every running application in Linux is represented as a process with different execution states.
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Why This Matters for Cloud & DevOps

Understanding Linux internals is essential because cloud infrastructure relies heavily on Linux servers.

These concepts help engineers:

- Deploy applications
 - Troubleshoot systems
 - Manage remote servers
 - Install software securely
 - Monitor running processes
 - Configure production environments
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Day 05 Summary

Today I explored the internal structure of Linux, including its architecture, file system, package management, hardware information, process states, and remote server management concepts. These topics provide the knowledge required to manage Linux-based cloud servers effectively and prepare for advanced DevOps tools such as Docker, Kubernetes, and AWS.

Status:  Day 05 Completed